

BIOCON FORMULATIONS	
MASTER PROOF OF APPROVED PRINTED ART WORK	VIAL LEAFLET
FOR COLOUR COPY	KIRSTY 100 U/ml, 10 ml
ANNEXURE-4 [Ref. No. : BM/QA/SOP/059]	FOR MALAYSIA
ARTWORK CODE:	BM/XXXX/XX



Kirsty™

Insulin Aspart Injection (rDNA origin) 100 U/mL

100 U/mL

COMPOSITION

Each mL contains:
Insulin Aspart Injection (rDNA origin) 100 U
Phenol 1.5 mg
Metacresol 1.72 mg (as preservative)

1 vial contains 10 mL equivalent to 1,000 units, 1 mL solution contains 100 units Insulin Aspart (equivalent to 3.5 mg).

Insulin Aspart is produced in *Pichia pastoris* by recombinant DNA technology. For the full list of excipients, see List of Excipients section.

PHARMACEUTICAL FORM

Solution for injection in a vial.
Clear, colourless, aqueous solution for injection in vial.

PHARMACOLOGICAL PROPERTIES*

Pharmacodynamics properties

Pharmacotherapeutic group:
Drugs used in diabetes. Insulins and analogues for injection, fast-acting, ATC code A10AB05.

* Unless otherwise specified; data presented in this P1 for Efficacy and Safety were referred from P1 of NovoRapid®

Mechanism of action

Kirsty™ produces a more rapid onset of action compared to soluble human insulin, together with a lower glucose concentration, as assessed within the first four hours after a meal. Kirsty™ has a shorter duration of action compared to soluble human insulin after subcutaneous injection.

When Kirsty™ is injected subcutaneously, the onset of action will occur within 10 to 20 minutes of injection. The maximum effect is exerted between 1 and 3 hours after injection. The duration of action is 3 to 5 hours.

Insulin aspart is equipotent to soluble human insulin on a molar basis.

Study MYL-1601D-3001 in T1DM patients to demonstrated equivalence in safety, immunogenicity and efficacy profile between Kirsty™ and NovoLog. 243 subjects in the Kirsty™ group and 242 subjects in the NovoLog group completed treatment. The treatment difference (SE) in TEAR responders between Kirsty™ and NovoLog was -2.86 and 90% CI were: -9.71, 3.99. Post-baseline, there were 24 subjects in Kirsty™ group and 35 subjects in the NovoLog group with newly positive ADA (treatment-induced TEAR), the difference was not statistically significant P-Value is 0.1348. The treatment differences (Kirsty™ – NovoLog) for HbA1c at Week 12 and Week 24 were 0.05 (95% CI: -0.05, 0.16) and 0.07 (95% CI: -0.06, 0.20), respectively.

Adults: clinical trials in patients with type 1 diabetes have demonstrated a lower postprandial blood glucose with Insulin Aspart compared to soluble human insulin. In two long-term open label trials in patients with type 1 diabetes comprising 1,070 and 584 patients, respectively, Insulin Aspart reduced glycated haemoglobin by 0.12 percentage points and by 0.15 percentage points compared to soluble human insulin; a difference of limited clinical significance.

Clinical trials in patients with type 1 diabetes have demonstrated a reduced risk of nocturnal hypoglycaemia with insulin aspart compared to soluble human insulin. The risk of daytime hypoglycaemia was not significantly increased.

Elderly: in a PK/PD trial the relative differences in the PD properties between insulin aspart and soluble human insulin in the elderly patients with type 2 diabetes were similar to those seen in healthy subjects and younger patients with diabetes.

Children and adolescents: when given to children, Insulin Aspart showed similar long-term glucose control compared to soluble human insulin. In clinical trials for children and adolescents aged 2 to 17, the pharmacodynamic profile of insulin aspart in children was similar to that seen in adults.

The efficacy and safety of Insulin Aspart given as bolus insulin in combination with either insulin detemir or insulin degludec as basal insulin have been studied for up to 12 months in two randomised controlled clinical trials in adolescents and children aged 1 to less than 18 years (n=712). The trials included 167 children aged 1–5 years, 260 aged 6–11 and 285 aged 12–17. The observed improvements in HbA_{1c} and the safety profiles were comparable between all age groups.

Pregnancy: a clinical trial comparing safety and efficacy of insulin aspart vs. soluble human insulin in the treatment of pregnant women with type 1 diabetes (322 exposed pregnancies) did not indicate any adverse effect of insulin aspart on pregnancy or on the health of the foetus/newborn.

In addition, the data from a clinical trial including 27 women with gestational diabetes randomised to treatment with insulin aspart vs. soluble human insulin showed similar safety profiles between treatments as well as a significant improvement in postprandial glucose control in the insulin aspart treated group.

Pharmacokinetic properties

Absorption, distribution and elimination

In Insulin Aspart substitution of amino acid proline with aspartic acid at position B28 reduces the tendency to form hexamers as observed with soluble human insulin. Insulin Aspart is therefore more rapidly absorbed from the subcutaneous layer compared to soluble human insulin.

The time to maximum concentration is, on average, half of that for soluble human insulin. A mean maximum plasma concentration of 492 pmol/l was reached 40 minutes after a subcutaneous dose of 0.15 U/kg bodyweight in type 1 diabetic patients. The insulin concentrations returned to baseline about 4 to 6 hours after dose. The absorption rate was somewhat slower in type 2 diabetic patients, resulting in a lower C_{max} (352 ± 240 pmol/l) and later t_{max} (60 minutes). The intra-individual variability in time to maximum concentration is significantly less for Insulin Aspart™ than for soluble human insulin, whereas the intra-individual variability in C_{max} for Insulin Aspart is larger.

Study MYL-1601D-1001 in healthy subjects to compare PK/PD between Kirsty™, NovoLog and NovoRapid has shown equivalence between 3 products was comparable. Geometric means for PK endpoints AUC_{0-∞}, C_{max} and C_{min} fell within the range of 0.8 to 1.25. The 90% CIs of MYL-1601D vs. NovoLog™ and the 95% CIs of MYL-1601D vs. NovoRapid for PD endpoints AUGR_{0-4h} and GRR_{0-4h} were within the prespecified range of 0.8 to 1.25.

Children and adolescents: the pharmacokinetic and pharmacodynamic properties of Insulin Aspart were investigated in children and adolescents with type 1 diabetes. Insulin aspart was rapidly absorbed in both age groups, with similar t_{max} as in adults. However, C_{max} differed between the age groups, stressing the importance of the individual titration of Insulin Aspart

Elderly: the relative differences in pharmacokinetic properties between insulin aspart and soluble human insulin in elderly patients with type 2 diabetes were similar to those observed in healthy subjects and in younger patients with diabetes. A decreased absorption rate was observed in elderly patients, resulting in a later t_{max} (82 minutes), whereas C_{max} was similar to that observed in younger patients with type 2 diabetes and slightly lower than in patients with type 1 diabetes.

Hepatic impairment: in patients with hepatic impairment, t_{max} was delayed to about 85 min, (50 min, in subjects with normal hepatic function) while AUC, C_{max} and CL/F were similar.

Renal impairment: a single dose pharmacokinetic study of insulin aspart in 18 subjects with normal to severely impaired renal function was performed. No apparent effect of creatinine clearance values on AUC, C_{max}, CL/F and t_{max} of insulin aspart was found. Data were limited in patients with moderate and severe renal impairment. Patients with renal failure necessitating dialysis treatment were not investigated

Pre-clinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity and toxicity to reproduction and development.

In *in vitro* tests, including binding to insulin and IGF-1 receptor sites and effects on cell growth, Kirsty™ behaved in a manner that closely resembled NovoRapid. Studies also demonstrate that the dissociation of binding to the insulin receptor of Kirsty™ is equivalent to NovoRapid.

The acute toxicology study with Biocon's Insulin Aspart in rats revealed no findings of concern doses up to approximately 80-fold the planned daily human dose. The only consistent observation was symptoms of hypoglycaemia. The 28-day repeat dose toxicology study in rats also found no signs of over toxicity at doses up to approximately 10-fold the anticipated daily human dose.

Like the acute toxicology study, the repeat dose study also demonstrated consistent observations of hypoglycaemia in both Biocon's Insulin Aspart and NovoRapid treated rats.

Hypoglycemia is an anticipated pharmacological effect and was not considered adverse for the purpose of these toxicology studies. In both acute repeat dose toxicity studies Kirsty™ was well tolerated.

CLINICAL PARTICULARS

Therapeutic indications

Kirsty™ is indicated for treatment of diabetes mellitus in adults, adolescents and children aged 1 year and above.

Posology and method of administration

Posology

Kirsty™ is a rapid-acting insulin analogue.

Kirsty™ dosage is individual and determined in accordance with the needs of the patient. It should normally be used in combination with intermediate-acting or long-acting insulin given at least once a day. Blood glucose monitoring and insulin dose adjustment are recommended to achieve optimal glycaemic control.

The individual insulin requirement in adults and children is usually between 0.5 and 1.0 U/kg/day. In a basal-bolus treatment regimen, 50–70% of this requirement may be provided by Kirsty™ and the remainder by intermediate-acting or long-acting insulin. Adjustment of dosage may be necessary if patients undertake increased physical activity, change their usual diet or during concomitant illness.

Kirsty™ has a faster onset and a shorter duration of action than soluble human insulin.

Due to the faster onset of action, Kirsty™ should generally be given immediately before a meal. When necessary Kirsty™ can be given soon after a meal.

Due to the shorter duration, Kirsty™ has a lower risk of causing nocturnal hypoglycaemic episodes.

Special populations

As with all insulin products, in elderly patients and patients with renal or hepatic impairment, glucose monitoring should be intensified and the insulin aspart dosage adjusted on an individual basis.

Paediatric population

Kirsty™ can be used in children and adolescents aged 1 year and above in preference to soluble human insulin when a rapid onset of action might be beneficial, for example, in the timing of the injections in relation to meals.

The safety and efficacy of Kirsty™ in children below 1 year of age have not been established. No data are available.

Transfer from other insulin products

When transferring from other insulin products, adjustment of the Kirsty™ dose and the dose of the basal insulin may be necessary.

Transfer from other insulin medicinal products

When transferring from other insulin medicinal products, adjustment of the Kirsty™ dose and the dose of the basal insulin may be necessary.

Interchangeability and Automatic Substitution

Kirsty™ has been developed as a similar biological medicinal product to NovoRapid and has been shown to have a comparable quality, safety and efficacy profile to NovoRapid. Therefore, interchangeability with the reference product NovoRapid may be considered if this is in agreement with the treating physician. However, automatic substitution (i.e. the practice by which a different product to that specified on the prescription is dispensed to the patient without the prior informed consent of the treating physician) and active substance-based prescription cannot apply to biologicals, including biosimilars. Such a differentiating approach towards biologicals ensures that treating physicians can make informed decision about treatments is in the interest of patient's safety.

Method of administration

Kirsty™ is administered subcutaneously by injection in the abdominal wall, the thigh, the upper arm, the deltoid region or the gluteal region. Injection sites should always be rotated within the same region in order to reduce the risk of lipodystrophy and cutaneous amyloidosis. As with all insulin products, subcutaneous injection in the abdominal wall ensures a faster absorption than other injection sites.

The duration of action will vary according to the dose, injection site, blood flow, temperature and level of physical activity. However, the faster onset of action compared to soluble human insulin is maintained regardless of injection site.

Instruction to be given to Patient on How to Handle Kirsty™ Vial

Before injecting this insulin,

1. Disinfect the rubber stopper with an alcohol swab.
2. Visually inspect the vial to ensure that there are no suspended impurities.
3. Draw air into the syringe, the same amount as the volume of insulin to be injected.
4. Inject the air into the vial: push the needle through the rubber stopper and press the plunger.
5. Turn the vial and syringe upside down.
6. Draw the correct dose of insulin into the syringe.
7. Before you take the needle out of the insulin vial, check the syringe for air bubbles. Make sure that there is no air left in the syringe: point the needle upwards and push the air out. Appearance of air bubble is a normal phenomenon, vigorous shaking immediately before the dose is administered may also result in the formation of air bubbles which could cause dosage errors; in that case tap the side of the syringe gently with your finger. A small air bubble may remain after tapping; this small air bubble will not affect your dose.
8. Check you have the right dose.
9. Inject straight away.

Continuous Subcutaneous Insulin Infusion (CSII):

Kirsty™ may be used for Continuous Subcutaneous Insulin Infusion (CSII) in pump systems suitable for insulin infusion. CSII should be administered in the abdominal wall. Infusion sites should be rotated. When used with an insulin infusion pump, Kirsty™ should not be mixed with any other insulin products. Patients using CSII should be comprehensively instructed in the use of the pump system and use the correct reservoir and tubing for the pump. The infusion set (tubing and cannula) should be changed in accordance with the instructions in the product information supplied with the infusion set.

Patients administering Kirsty™ by CSII must have alternative insulin delivery method available in case of pump system

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CARTON CLOSURE PACK - A

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SENSO AVANZAM.
FEEDING WAY

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26/02/2013

26/02/2013

MAGAZINE FIXED SIDE

Leaflet size to be same as
of 10ml size: 180x110 mm.

swatch
I will
check and
report back 26/02/2013

	A	B	OPEN SIZE
3 ml	20	90	228x180
5 ml	23	100	180x100
10 ml	23	110	180x110
MYOKINASE 10ml	23	110	180x110

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DATE
25/02/2013

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